Batterjee Medical College
For Science and Technology

Fever of Unknown Origin (FUO) & Travel Medicine (Pt. 1)

An approach to a Case of Fever

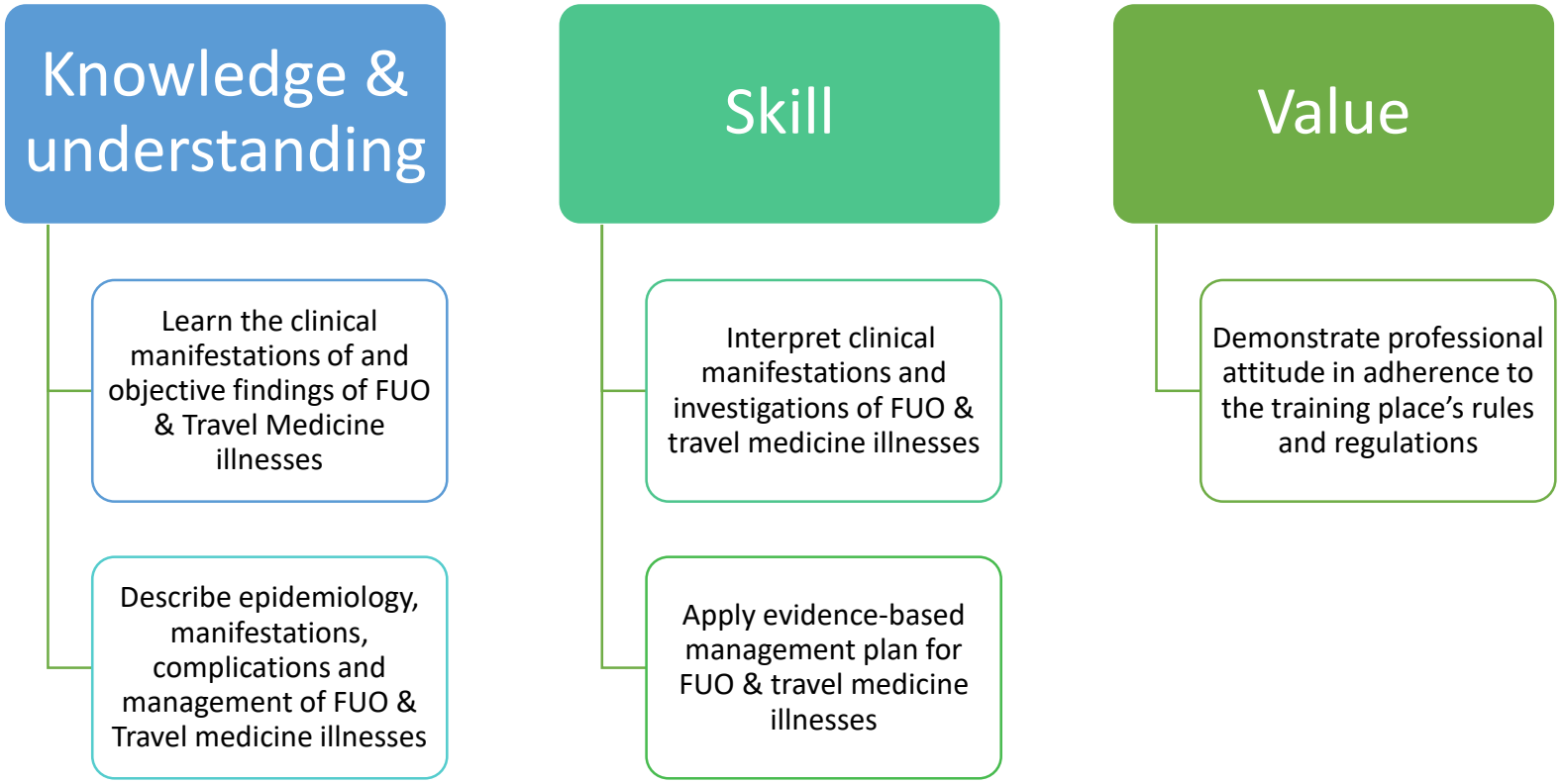
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BATTERJEE MEDICAL COLLEGE

BMC
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- Fever of Unknown Origin
- Malaria
- Typhoid & Paratyphoid
- Traveler's Diarrhea

Learning Objectives



The Case

A 29-year-old woman comes to the physician because of high-grade fever, headache, and muscle pain. A week ago, she returned from India, where she was working with an NGO to build houses in rural areas. She took a single dose of chloroquine 1 week before her trip and once-weekly thereafter. Her last dose was 2 days ago. She has no history of a serious illness. Her temperature is 39.3°C (102.7°F), pulse is 102/min, respirations are 22/min, and blood pressure is 112/78 mm Hg. Physical examination shows jaundice. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Her hemoglobin concentration is 10 g/dL. A peripheral blood smear shows fragmented erythrocytes and occasional ring-shaped inclusion bodies within RBCs. Which of the following is the most likely cause of infection in this patient?

a. *Plasmodium ovale*
b. *Plasmodium falciparum*
c. Dengue virus
d. Chikungunya virus
e. Hepatitis A virus

Keep this case in mind.

We'll build the framework, then come back and solve it together at the end.

What does the blood smear tell you about which organism this can't be?

FUO - Overview



Classic FUO: temperature > 38.3°C (100.1°F) on multiple occasions, lasting > 3 weeks, with no clear etiology despite 3 outpatient visits or 3 days in hospital.

But "classic" FUO is only one of four recognized categories — each with its own definition and differential:

Nosocomial FUO

Hospitalized patient, no infection present/incubating at admission; unexplained after 3 days of evaluation

Neutropenic FUO

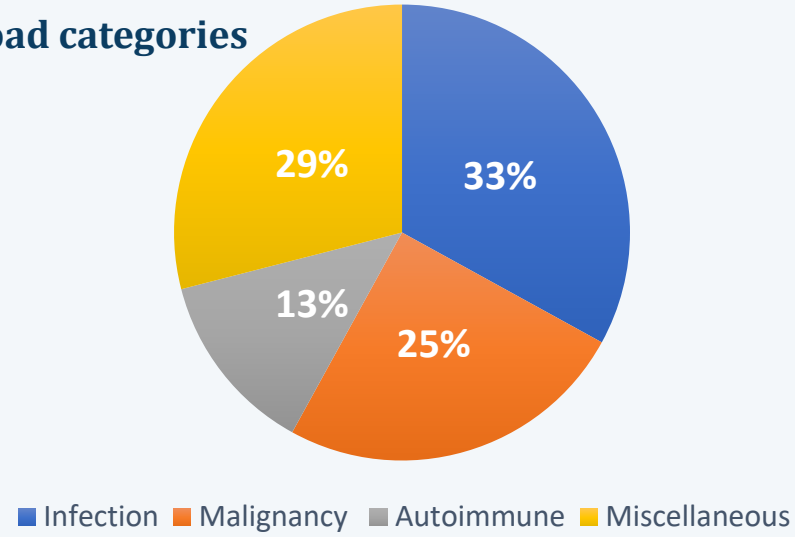
ANC < 500/mm³ (or expected to fall below); unexplained after 3 days, including 2 days of culture incubation

HIV-associated FUO

Confirmed HIV infection; fever > 4 weeks (outpatient) or > 3 days (inpatient), unexplained after appropriate workup

FUO - Etiology

Broad categories



Proportions vary widely by study population, region, and era — treat these as rough

Healthcare-associated FUO

- Drug fever
- Intravascular catheter-related infection
- Venous thromboembolism
- Clostridioides difficile colitis
- Occult abscess; transfusion reactions
- Candidemia (if critically ill)

Infection

TB, brucellosis, UTI, endocarditis, HIV, sinusitis, malaria

Malignancy

Lymphoma & leukemia, renal cell carcinoma

Inflammatory

Adult-onset Still disease, SLE, stroke

FUO - Etiology

TABLE 17-1 Etiology of Fever of Unknown Origin (FUO) Over the Past 25 Years: Findings from Large FUO Studies

FIRST AUTHOR (COUNTRY, YEAR OF PUBLICATION)	NO. OF PATIENTS (RECRUITMENT PERIOD)	PERCENTAGE OF CASES DUE TO INDICATED CAUSE				
		INFECTIONS	NONINFECTIOUS INFLAMMATORY DISEASES	NEOPLASMS	MISCELLANEOUS	UNKNOWN
Western Countries						
De Kleijn et al. (Netherlands, 1997)	167 (1992–1994)	26	24	13	8	30
Vanderschueren et al. (Belgium, 2003)	185 (1990–1999)	11	18	10	8	53
Hot et al. (France, 2005)	280 (1995–2005)	11	20	27	9	33
Zenone et al. (France, 2006)	144 (1999–2005)	23	26	10	15	26
Bleeker-Rovers (Netherlands, 2007)	73 (2003–2005)	16	22	7	4	51
Mansueto et al. (Italy, 2008)	91 (1991–2002)	32	12	14	10	32
Vanderschueren et al. (Belgium, 2009)	114 (2003–2007)	15	22	13	10	40
Efstathiou et al. (Greece, 2010)	112 (2001–2007)	30	33	11	5	21
Pedersen et al. (Denmark, 2012)	52 (2005–2010)	19	33	8	0	40
Robine et al. (France, 2014)	103 (2002–2012)	12	30	3	5	51
Vanderschueren et al. (Belgium, 2014)	436 (2000–2010)	17	24	11	10	39
Total	**1757**	**19**	**24**	**12**	**8**	**38**

TABLE 17-1 Etiology of Fever of Unknown Origin (FUO) Over the Past 25 Years: Findings from Large FUO Studies (Continued)

FIRST AUTHOR (COUNTRY, YEAR OF PUBLICATION)	NO. OF PATIENTS (RECRUITMENT PERIOD)	PERCENTAGE OF CASES DUE TO INDICATED CAUSE				
		INFECTIONS	NONINFECTIOUS INFLAMMATORY DISEASES	NEOPLASMS	MISCELLANEOUS	UNKNOWN
Ergonul et al. (Turkey, 2005)	80 (1993–1999)	52	16	18	3	11
Brahim et al. (Turkey, 2005)	97 (1990–2005)	36	8	16	5	35
Chin et al. (Taiwan, 2006)	94 (2001–2002)	57	7	9	9	18
Colpan et al. (Turkey, 2007)	71 (2001–2004)	45	27	14	6	9
Hu et al. (China, 2008)	142 (2002–2003)	36	32	13	5	14
Kucukardali et al. (Turkey, 2008)	154 (2003–2004)	34	31	14	5	16
Ali-Eldin et al. (Egypt, 2011)	93 (2009–2010)	42	15	30	0	12
Bandyopadhya et al. (India, 2011)	164 (2008–2009)	55	11	22	0	12
Mete et al. (Turkey, 2012)	100 (2001–2009)	26	38	14	2	20
Ma et al. (China, 2012)	397 (2000–2009)	49	18	16	7	10
Ryuko et al. (Japan, 2013)	174 (2004–2010)	41	27	7	6	19
Mahmood et al. (Pakistan, 2013)	205 (2006–2011)	49	20	13	2	17
Alvi et al. (Iran, 2013)	106 (2007–2011)	44	18	12	10	15

FUO – Clinical Features

Manifestations that may accompany fever, without pointing to one specific entity

Chills

A subjective sensation of cold, often with shivering

Rigors

A severe form of chills, with pronounced shivering and teeth chattering

Night sweats

Drenching sweats that soak clothing or bedding

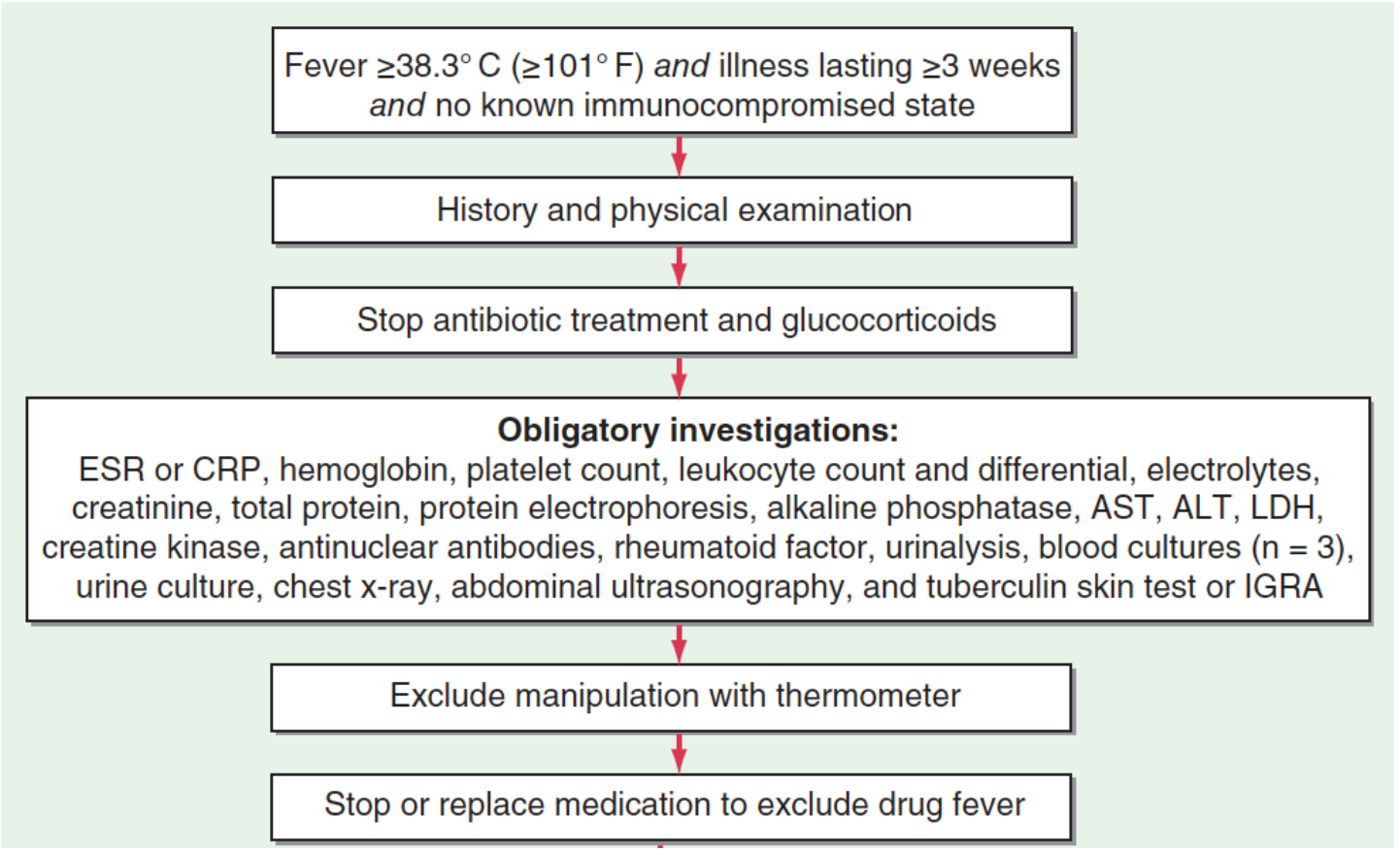
Change in mental status

Especially notable at the extremes of age

Pearl: Also look for systemic clues of the underlying cause: skin changes, constitutional symptoms, anemia, unintentional weight loss.

FUO - Patient Approach

1. **History & physical**
2. **Stop confounders**
3. **Obligatory laboratory workup**



Pearl: Exception: don't withhold empiric therapy if there's rapid clinical deterioration or a life-threatening etiology is suspected.

FUO - Patient Approach

Advanced imaging

PET-CT, scintigraphy, and similar functional imaging can localize an occult inflammatory or malignant focus when standard workup is unrevealing — particularly useful for large-vessel vasculitis and occult malignancy.

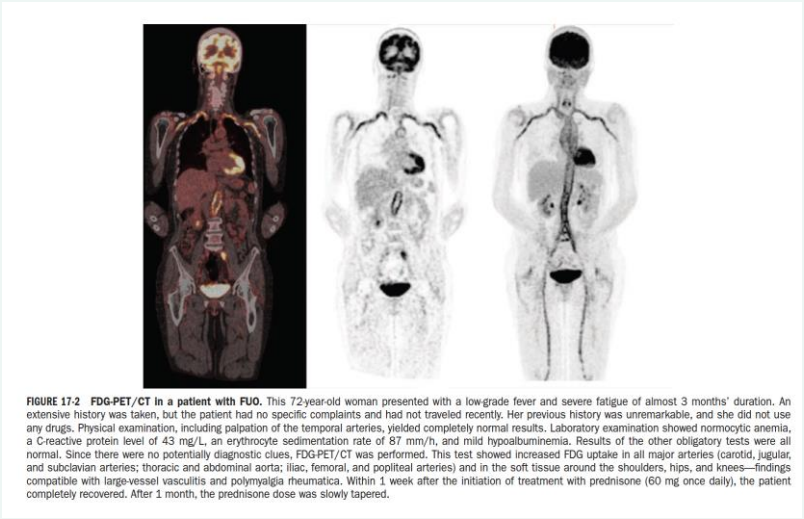
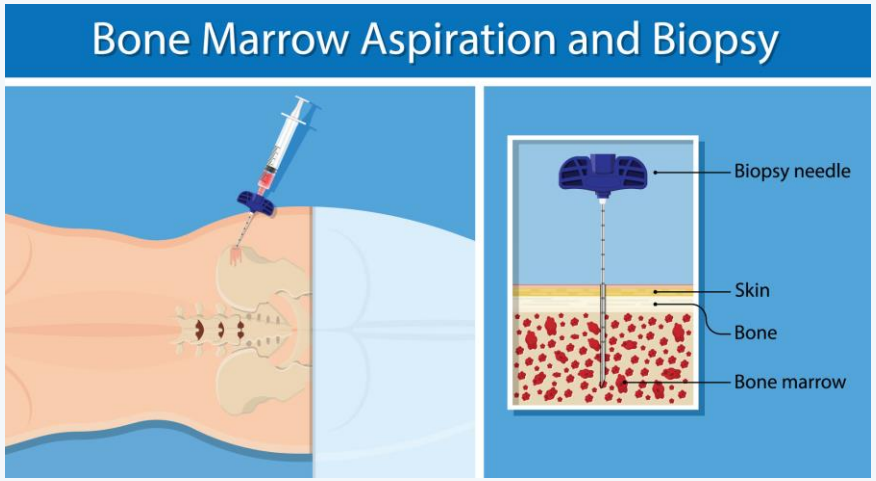


FIGURE 17-2 FDG-PET/CT in a patient with FUO. This 72-year-old woman presented with a low-grade fever and severe fatigue of almost 3 months' duration. An extensive history was taken, but the patient had no specific complaints and had not traveled recently. Her previous history was unremarkable, and she did not use any drugs. Physical examination, including palpation of the temporal arteries, yielded completely normal results. Laboratory examination showed normocytic anemia, a C-reactive protein level of 43 mg/L, an erythrocyte sedimentation rate of 87 mm/h, and mild hypoalbuminemia. Results of the other obligatory tests were all normal. Since there were no potentially diagnostic clues, FDG-PET/CT was performed. This test showed increased FDG uptake in all major arteries (carotid, jugular, and subclavian arteries; thoracic and abdominal aorta; iliac, femoral, and popliteal arteries) and in the soft tissue around the shoulders, hips, and knees—findings compatible with large-vessel vasculitis and polymyalgia rheumatica. Within 1 week after the initiation of treatment with prednisone (60 mg once daily), the patient completely recovered. After 1 month, the prednisone dose was slowly tapered.

Invasive procedures

- Biopsy of lymph node, bone marrow, or other tissue when a tumor or abscess is highly suspected
- Analysis and culture of any abnormal fluid collection (e.g., joint, pleural)



Pearl: Reserve invasive testing for high pre-test suspicion — it's a second-tier step, not a routine reflex.

FUO - Management

Avoid empiric therapy

Antibiotics and glucocorticoids can mask the underlying disease and delay diagnosis

Know the exception

Treat empirically only with rapid deterioration or a suspected life-threatening cause

Treat the cause once found

Manage according to the specific diagnosis identified

If it remains undiagnosed

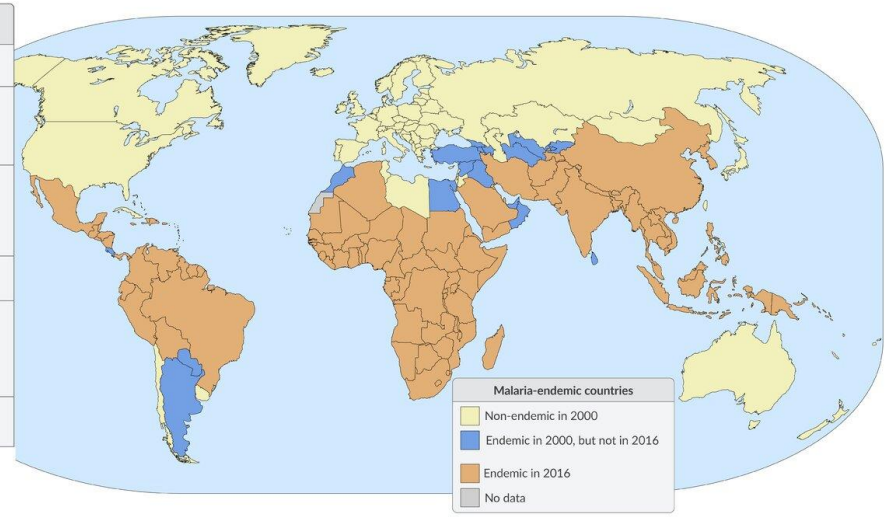
Persisting FUO despite advanced workup warrants specialist consultation

Pearl: A specialist may recommend a diagnostic trial of anti-TB therapy, colchicine, steroids, or anakinra (an IL-1 receptor antagonist) when advanced diagnostics remain unrevealing.

Malaria Overview

A protozoal infection transmitted via the bite of an infected female Anopheles mosquito. Prevalent across tropical and subtropical regions — especially sub-Saharan Africa, South and Southeast Asia, and parts of the Middle East and South America.

Malaria *Several species of Plasmodium*	
Transmission	- Female mosquitoes of the Anopheles genus
Incubation period	**Depends on parasite** - Falciparum malaria (P. falciparum): 7–15 days - Tertian malaria (P. vivax and P. ovale): 12–18 days - Quartan malaria (P. malariae): 18–40 days - Significantly longer incubation periods may occur in some cases
Clinical features	**Flu-like symptoms, headache, bodyache, fever** - Falciparum malaria: irregular bouts of fever, life-threatening complications are frequent - Tertian malaria: fever occurring every 48 hours - Quartan malaria: fever occurring every 72 hours
Rates	- New cases per year: 212 million - Deaths per year: 429,000
Prevention	**Prophylaxis** - Insect repellents - Mosquito-proof housing - Insecticide-treated bed nets - Chemoprophylaxis before travel to endemic areas
Note	- No vaccine available - Resistance is increasing - Incidence and case fatality is decreasing worldwide



~212 million

new malaria cases estimated per year, worldwide

~429,000

malaria deaths estimated per year

No vaccine

widely deployed historically; resistance to antimalarials is increasing

Malaria - Etiology

- caused by one of five plasmodium parasites:



Classic disease and fever patterns of different plasmodium species [9][10].		
Different species of plasmodium [11][12]	**Disease**	**Fever spikes**
P. vivax **P. ovale**	• Tertian malaria (usually less severe)	• Every 48 hours
P. malariae	• Quartan malaria (usually less severe)	• Every 72 hours
P. falciparum	• Falciparum malaria (most severe form; also known as malignant tertian malaria)	• Irregular
P. knowlesi	• Quotidian malaria	• Every 24 hours

Pearl: P. vivax and P. ovale carry a dormant liver stage (hypnozoites) that the other species lack — the key to understanding relapse, coming up next.

Malaria - Pathogenesis

1 Inoculation

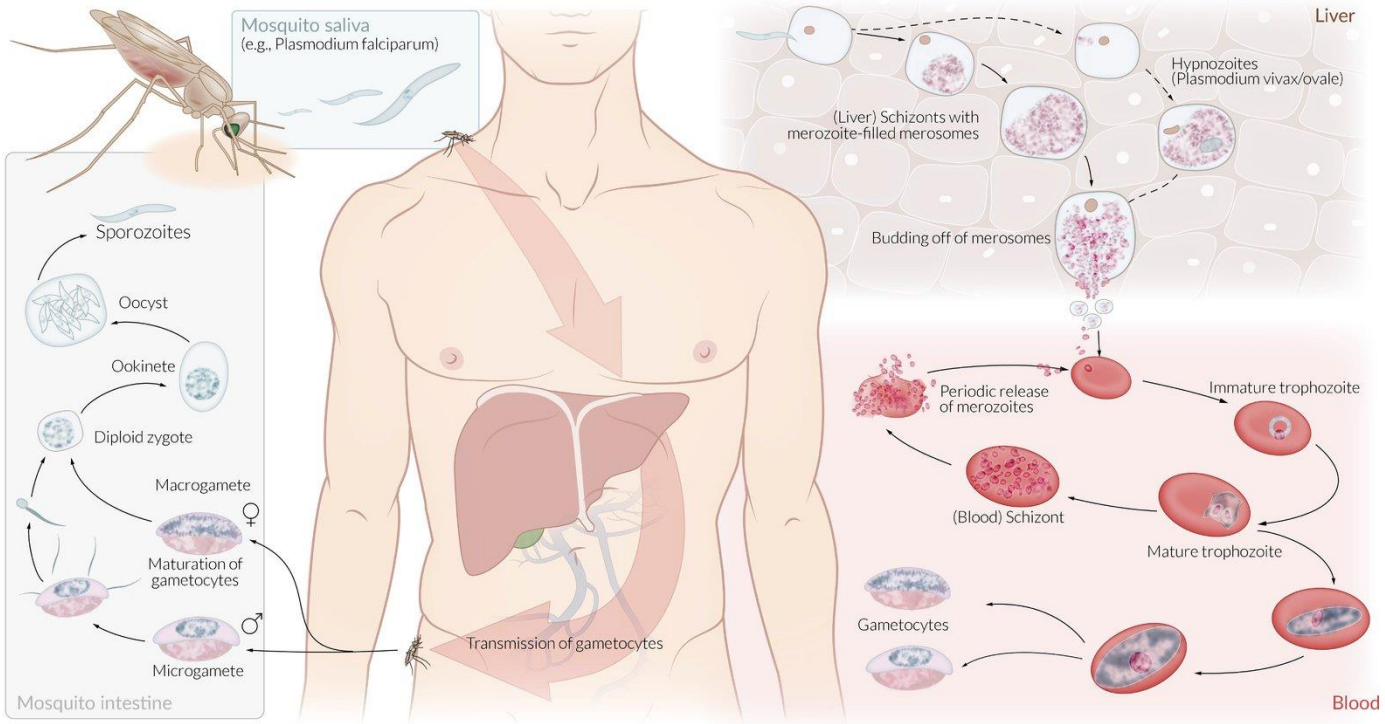
An infected Anopheles mosquito injects sporozoites into the bloodstream

2 Liver stage

Sporozoites invade hepatocytes, multiply asexually, and form schizonts full of merozoites

3 Blood stage

Merozoites are released and invade red blood cells, maturing into trophozoites and RBC schizonts



Malaria - Pathogenesis

4 Hemolysis & symptoms

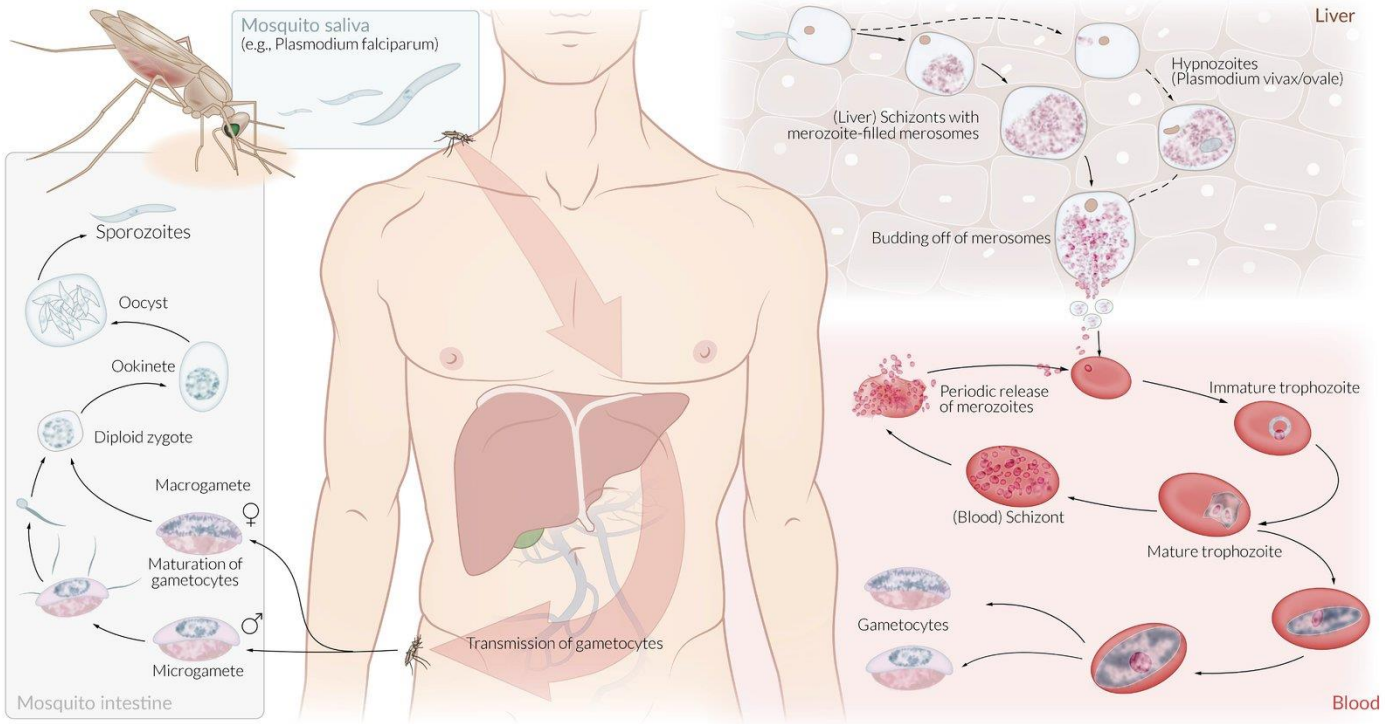
RBC rupture releases more merozoites — this cyclical rupture causes the fever and hemolysis of malaria

5 Gametocyte formation

Some merozoites differentiate into sexual-stage gametocytes instead of continuing the blood cycle

6 Back to the mosquito

A new mosquito ingests gametocytes, which mature and migrate to its salivary glands to infect the next host



Malaria - Relapse vs. Recrudescence vs. Reinfection

Why symptoms can return long after apparent cure

Relapse

Dormant liver hypnozoites reactivate and re-seed the blood

Species: P. vivax and P. ovale only

Requires primaquine (or tafenoquine) for radical cure — check G6PD first

Recrudescence

Blood-stage parasites persisted at sub-microscopic levels and rebound

Species: Any species; classically P. falciparum or P. malariae

Suggests inadequate treatment or drug resistance

Reinfection

An entirely new infection from a new mosquito bite

Species: Any species, with new exposure

Distinguished by travel/exposure timeline, not by testing alone

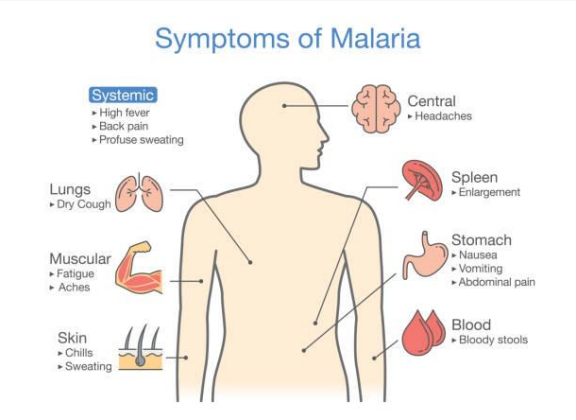


Pearl: A patient successfully treated for malaria months ago who now has the same illness after no new travel: think relapse (hypnozoite reactivation), not treatment failure.

Malaria - Clinical Features

Symptoms

- Fever (can be cyclical)
- Chills, malaise, headache
- Myalgias
- Gastrointestinal symptoms



Why the hemolysis?

Each cycle of RBC rupture releases new merozoites — the repeated destruction of parasitized red cells is what drives the anemia and jaundice.

Signs

- Hemolytic anemia — indirect hyperbilirubinemia, hemoglobinuria
- Thrombocytopenia
- Splenomegaly
- Renal failure
- Disseminated intravascular coagulation (DIC)

Malaria - Severe (Complicated) Disease

Almost always P. falciparum — mature schizonts bind vascular endothelium, causing capillary obstruction

Cerebral malaria

Seizures, coma

Pulmonary edema

Can progress to ARDS

Renal failure

Nephritis, acute kidney injury

Severe anemia

From high rates of parasitized RBCs

Hypoglycemia

Often missed — check glucose

Shock / DIC

Marks critical, life-threatening disease

Pearl: Any of these findings in a patient with malaria is a medical emergency — treat as severe disease and consider ICU-level care.

Malaria - Diagnostics

To evaluate organ dysfunction

- CBC
- Comprehensive metabolic panel
- LFTs
- Coagulation panel

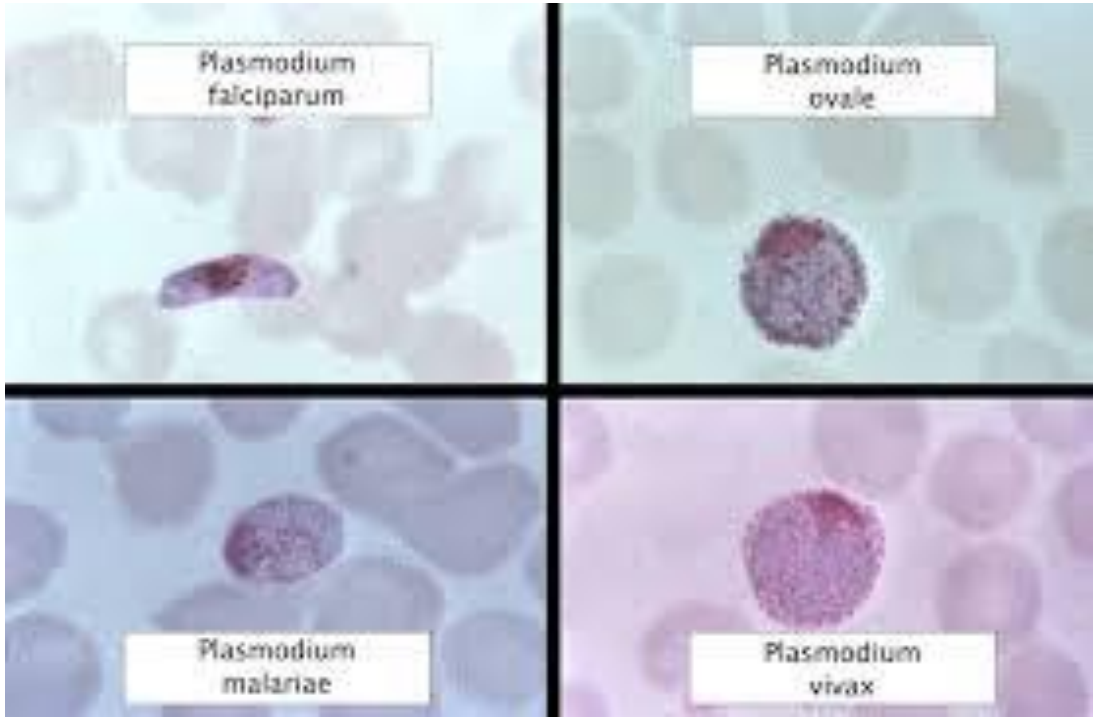
Parasitological testing

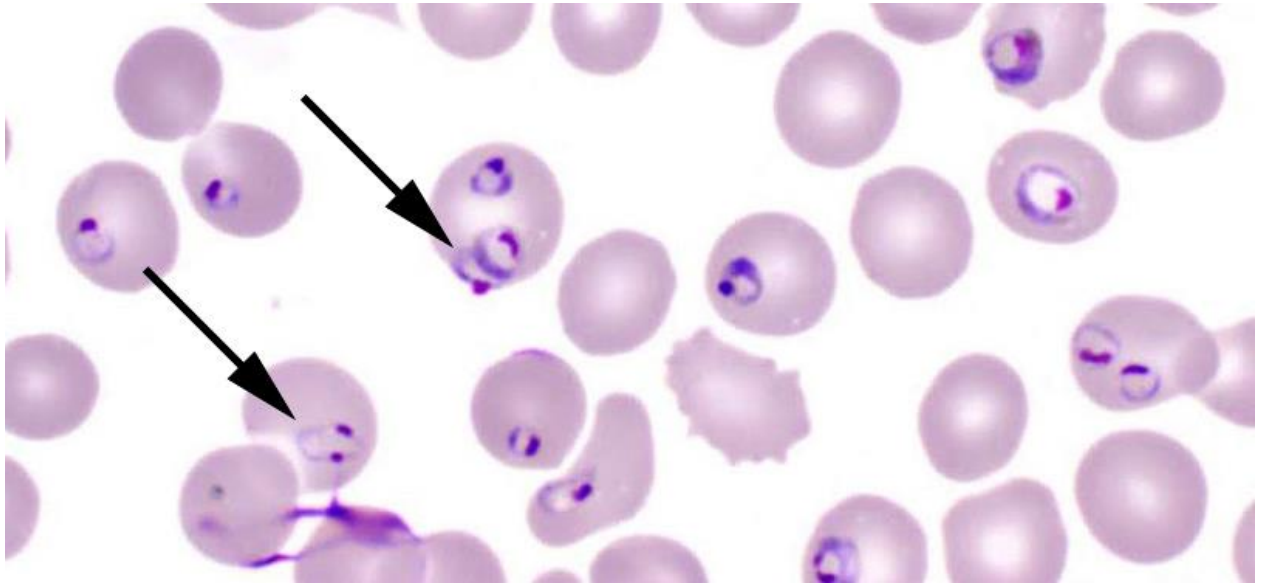
- Thick & thin blood smear (Giemsa/Wright stain) — the gold standard
- Rapid antigen detection test — faster, less sensitive

Schuffner granules (fine brick-red dots) within RBCs infected with P. vivax and P. ovale; **crescent-shaped gametocytes** are characteristic of P. falciparum.

Don't stop at one: a single negative smear does not rule out malaria — repeat every 12–24 hours for a total of 3 sets before excluding the diagnosis.

Malaria - Diagnostics





Malaria - Management

P. vivax, P. ovale, P. malariae

- Treat with chloroquine
- For P. vivax and P. ovale, add primaquine to eradicate dormant hypnozoites and prevent relapse
- Check G6PD level first — primaquine can precipitate severe hemolysis in G6PD deficiency

P. falciparum

- Assume chloroquine resistance
- Options: quinine + doxycycline; quinine + sulfadoxine/pyrimethamine; mefloquine; atovaquone/proguanil
- Severe disease: IV artesunate (artemisinin therapy) is the drug of choice

Pearl: Monitor treatment with repeat smears at 48 hours, expecting a > 75% drop in parasitemia; consider exchange transfusion if parasitemia exceeds 10–15%.

Malaria – Prevention

Bite avoidance first: bed nets, window screens, insecticides, and repellents with 30–35% DEET.

Destination	Recommended chemoprophylaxis
Central America, Haiti, parts of the Middle East	Chloroquine (still effective in these regions)
Most other endemic areas	Mefloquine or atovaquone/proguanil
Southeast Asia	Doxycycline or atovaquone/proguanil (resistance to other agents is common)

Side effects can drive the choice

- Atovaquone/proguanil: avoid in renal disease and pregnancy
- Mefloquine: avoid with seizures or psychiatric conditions
- Chloroquine: well tolerated, safe in pregnancy
- Primaquine: contraindicated in G6PD deficiency (hemolysis risk)

Malaria – Prevention

Centers for Disease Control and Prevention
CDC 24/7: Saving Lives, Protecting People™

A-Z Index

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Advanced Search

Travelers' Health

COVID-19 Travel Health Information

- International Travel for U.S. Citizens
- International Travel for Non-Citizens
- Testing Requirement
- Vaccination and Travel
- Travel in the U.S.
- Cruise Travel

Travel Assessment

A tool to help you know the requirements to board a flight to the United States.

Get Started

Destinations



Where are you going?

Saudi Arabia

Go

View all destinations

Typhoid & Paratyphoid - Overview



Caused by ***Salmonella enterica***, serotype Typhi and Paratyphi (A, B, and C). Most prevalent in resource-limited regions with poor sanitation: East and Southeast Asia, Africa, and Central and South America.



Incubation

5–30 days



Transmission

Fecal-oral route — contaminated food and water

Typhoid & Paratyphoid - Clinical Features

Core features

- Fever, often stepwise/escalating over days
- Relative bradycardia (pulse-temperature dissociation) — a classic finding
- Constipation or diarrhea
- Headache
- Coated tongue



Additional findings

- Rose spots — a small, speckled, rose-colored exanthem on the lower chest and abdomen
- Nonspecific abdominal pain
- Diarrhea
- Neurological symptoms in severe cases: delirium, coma

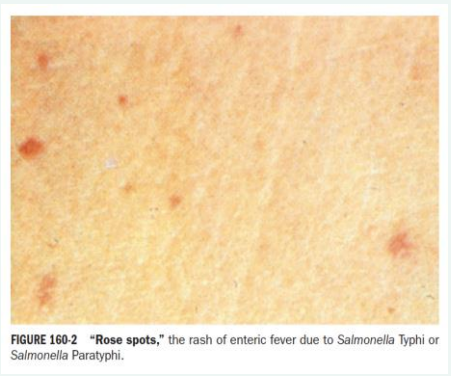


FIGURE 160-2 **"Rose spots,"** the rash of enteric fever due to *Salmonella* Typhi or *Salmonella* Paratyphi.



Typhoid & Paratyphoid – Diagnostics & Management

Diagnosis

- Blood cultures — most sensitive in the first week of illness
- Stool and urine cultures — more often positive later in the illness

Treatment

- Ciprofloxacin or azithromycin
- A cephalosporin (e.g., ceftriaxone) if quinolone resistance is present
- Symptomatic and supportive care

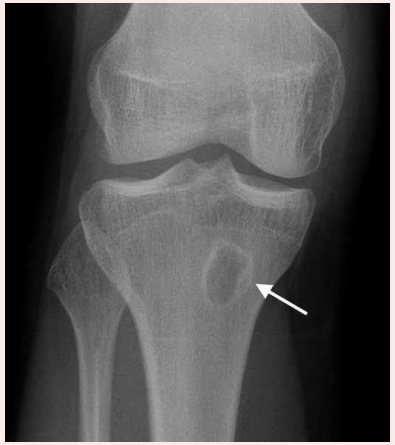
TABLE 160-1 Antibiotic Therapy for Enteric Fever in Adults

INDICATION	AGENT	DOSAGE (ROUTE)	DURATION, DAYS
Empirical Treatment			
	Ceftriaxone[a]	2 g/d (IV)	10–14
	Azithromycin[b]	1 g/d (PO)	5
Fully Susceptible			
Optimal treatment	Ciprofloxacin[c]	500 mg bid (PO) or 400 mg q12h (IV)	5–7
	Azithromycin	1 g/d (PO)	5
Alternative treatment	Amoxicillin	1 g tid (PO) or 2 g q6h (IV)	14
	Chloramphenicol	25 mg/kg tid (PO or IV)	14–21
	Trimethoprim-sulfamethoxazole	160/800 mg bid (PO)	7–14
Multidrug-Resistant			
Optimal treatment	Ceftriaxone[a]	2 g/d (IV)	10–14
	Azithromycin	1 g/d (PO)	5
Alternative treatment	Ciprofloxacin	500 mg bid (PO) or 400 mg q12h (IV)	5–14
Quinolone-Resistant			
Optimal treatment	Ceftriaxone	2 g/d (IV)	10–14
	Azithromycin	1 g/d (PO)	5
Alternative treatment	High-dose ciprofloxacin[d]	750 mg bid (PO) or 400 mg q8h (IV)	10–14

Typhoid — Complications & the Carrier State

Complications (untreated, week 3+)

- Intestinal hemorrhage and perforation — from necrosis of Peyer's patches in the terminal ileum; presents as an acute abdomen
- Typhoid encephalopathy, myocarditis, cholecystitis
- Osteomyelitis — classically associated with underlying sickle cell disease



Chronic carrier state

- About 1–5% of patients become chronic carriers, with the organism persisting in the gallbladder (especially with gallstones)
- Defined as fecal shedding for more than 1 year
- Historically illustrated by the case of “Typhoid Mary”
- Treatment: a prolonged antibiotic course; cholecystectomy if refractory



Traveler's diarrhea

Definition: sudden onset of ≥ 3 loose or watery stools per day, plus at least one additional GI or systemic symptom (fever, cramps, nausea, abdominal pain, or blood in the stool).



Onset: usually within 1 week of arrival; watery diarrhea lasting 2–4 days.

Etiology

Enterotoxigenic E. coli	*> 50% of cases*
Campylobacter, Shigella, Salmonella	
Parasites: Giardia, Entamoeba, Cryptosporidium	

Pearl: Dysentery — bloody diarrhea with fever — points toward Shigella or Entamoeba rather than the typical toxin-mediated picture.

Traveler's diarrhea

Risk factors & diagnostics

- Highest risk: South and Southeast Asia, sub-Saharan Africa, the Middle East, and Latin America
- Risk also depends on trip type/duration and host factors
- Identifying the specific organism is generally unnecessary — microbiologic workup is reserved for systemic illness or prolonged disease

Management

- Fluid replacement — the mainstay of treatment
- Antibiotics have a limited role: fluoroquinolone, azithromycin, or rifaximin for more severe cases
- Symptomatic therapy: antimotility agents such as loperamide or diphenoxylate

Approach to Fever

Key History:

- Severity
- Duration
- Pattern
- associated localizing symptoms such as rash, wheezing, cough, and ear discharge, etc.
- Night sweats, weight loss
- Travel history
- Contact with sick persons
- Contact with animals
- Immunizations
- Recent surgery
- Medications

Approach to Fever

Key Physical:

- Vital signs
- HEENT
- Neck
- Heart
- Lung
- Abdominal
- Skin

Approach to Fever

DDx:

Infections
Tuberculosis (especially extrapulmonary)
Abdominal abscesses
Pelvic abscesses
Dental abscesses
Endocarditis
Osteomyelitis
Sinusitis
Cytomegalovirus
Epstein-Barr virus
Human immunodeficiency virus
Lyme disease
Prostatitis
Sinusitis
Malignancies
Chronic leukemia
Lymphoma
Metastatic cancers
Renal cell carcinoma
Colon carcinoma
Hepatoma
Myelodysplastic syndromes
Pancreatic carcinoma
Sarcomas
Sarcoidosis

Autoimmune conditions
Adult Still's disease
Polymyalgia rheumatica
Temporal arteritis
Rheumatoid arthritis
Rheumatoid fever
Inflammatory bowel disease
Reiter's syndrome
Systemic lupus erythematosus
Vasculitides
Miscellaneous
Drug-induced fever
Complications from cirrhosis
Factitious fever
Hepatitis (alcoholic, granulomatous, or lupoid)
Deep venous thrombosis

Approach to Fever

General Workup:

- CBC
- BMP
- LFTs
- UA
- Urine culutres
- CXR
- Sputum Cultures
- Stool analysis
- Viral antibodies

Specific Workup:

- ANA
- RF
- LP
- Serology
- Abdominal Imaging
- TB

To Summarize

	Cause	Hallmark	Treatment
Malaria	Plasmodium spp. via Anopheles bite	Cyclical fever, hemolytic anemia, splenomegaly	Species-specific antimalarials; artesunate for severe disease
Typhoid fever	Salmonella Typhi/Paratyphi	Stepwise fever, relative bradycardia, rose spots	Ciprofloxacin or azithromycin
Traveler's diarrhea	Enterotoxigenic E. coli (mostly)	Watery diarrhea within 1 week of arrival	Fluids ± antibiotic ± loperamide

Take-home points

- Only P. vivax and P. ovale relapse (dormant hypnozoites) — always check G6PD before primaquine
- A single negative malaria smear doesn't exclude the diagnosis — repeat it
- Rose spots + relative bradycardia + travel to an endemic area = think typhoid

Case Scenarios



Case Scenario 1

A 29-year-old woman comes to the physician because of high-grade fever, headache, and muscle pain. A week ago, she returned from India, where she was working with an NGO to build houses in rural areas. She took a single dose of chloroquine 1 week before her trip and once-weekly thereafter. Her last dose was 2 days ago. She has no history of a serious illness. Her temperature is 39.3°C (102.7°F), pulse is 102/min, respirations are 22/min, and blood pressure is 112/78 mm Hg. Physical examination shows jaundice. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Her hemoglobin concentration is 10 g/dL. A peripheral blood smear shows fragmented erythrocytes and occasional ring-shaped inclusion bodies within RBCs. Which of the following is the most likely cause of infection in this patient?

a. *Plasmodium ovale*
b. *Plasmodium falciparum*
c. Dengue virus
d. Chikungunya virus
e. Hepatitis A virus

Case Scenario 2

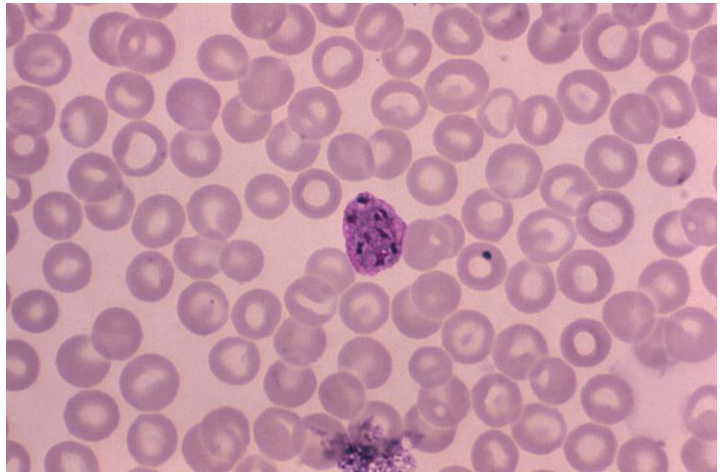
A 14-year-old girl is brought to the physician by her father because of fever, chills, abdominal pain, and profuse non-bloody diarrhea. Her symptoms began 1 week ago when she had several days of low-grade fever and constipation. She returned from Indonesia 2 weeks ago, where she spent the summer with her grandparents. Her temperature is 39.3°C (102.8°F). Examination shows diffuse abdominal tenderness and mild hepatosplenomegaly. There is a faint salmon-colored maculopapular rash on her trunk and abdomen. Which of the following is the most likely causal organism?

a. *Escherichia coli*
b. *Clostridioides difficile*
c. *Shigella dysenteriae*
d. *Salmonella typhi*

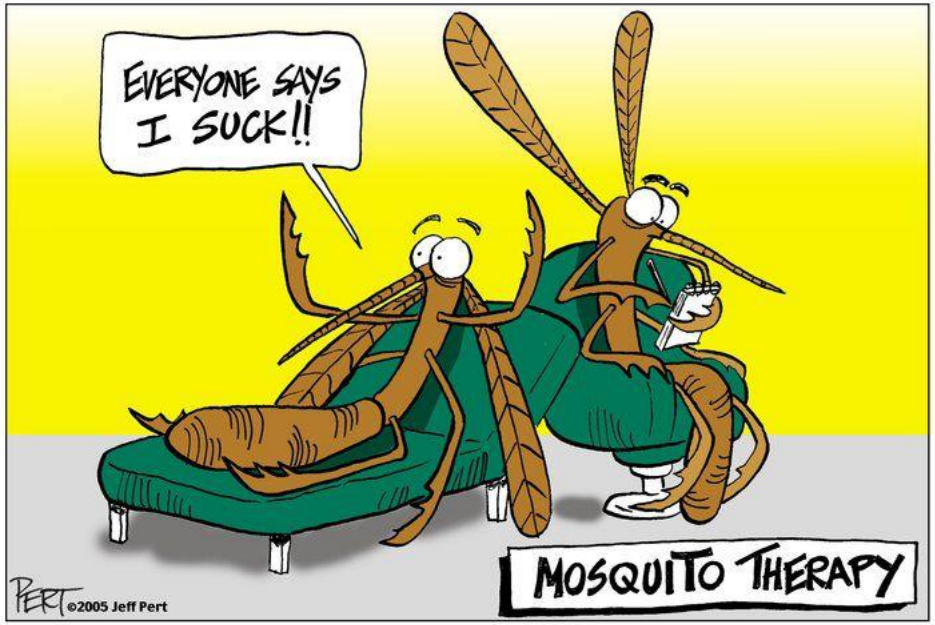
Case Scenario 3

A 55-year-old woman comes to the physician because of fever, chills, headache, and nausea over the past 3 days. Nine months ago, she returned from a vacation in Indonesia where she had experienced similar symptoms and episodic fever. She was treated with chloroquine and recovered uneventfully. Her temperature is 39.1°C (102.4°F), pulse is 97/min, and blood pressure is 123/85 mm Hg. Physical examination shows scleral icterus. The abdomen is soft; bowel sounds are active. Neurologic examination is unremarkable. Her hemoglobin concentration is 10 g/dL. A photomicrograph of a peripheral blood smear is shown. Which of the following is the most likely cause of the recurrence of symptoms in this patient?

a. Decline in circulating antibodies
b. Reinfection by Anopheles mosquito
c. Natural drug resistance
d. Reactivation of dormant liver stage
e. Dissemination within macrophages



Questions



References & further reading

- Harrison's Principles of Internal Medicine 20th edition
- Amboss
- 1st aid for the internal medicine boards, 4th edition
- Step-up to medicine 5th edition

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Thank You